# Summary of activities, 9 March 2026

## 1. PD07 gDNA extraction and primer validation

On Monday (09Mar26), I prepared 500-$\mu$L 10$\mu$M stock solutions of the *GBA1* L444P and N370S primer pairs using nuclease-free water as a diluent; these are stored in the 4°C in a box with my name on it in red tape. The math for the dilutions is attached.

On Tuesday, I harvested one dish (#4) of PD07 fibroblasts and extracted gDNA, yielding a concentration of 27.3 ng/$\mu$L. According to Tanner, this is in line with his typical yields, although higher is preferred. I stored the gDNA pellet in the −20°C in a box with my name on it; the tube is labeled `PD07-FB P4 gDNA 10Mar26`. 4$\mu$L were then used for PCR to confirm that the primers work; I also used 4$\mu$L of the HC2-FB gDNA pellet provided to me by Tanner. The PCR gel confirms amplicons migrating around 300 and 400 bp, as expected (see attached). I will provide the gDNA to Angel (David Liu Lab) on Monday at 2:00PM at the Broad so that he can sequence our samples.

## 2. Cell culture

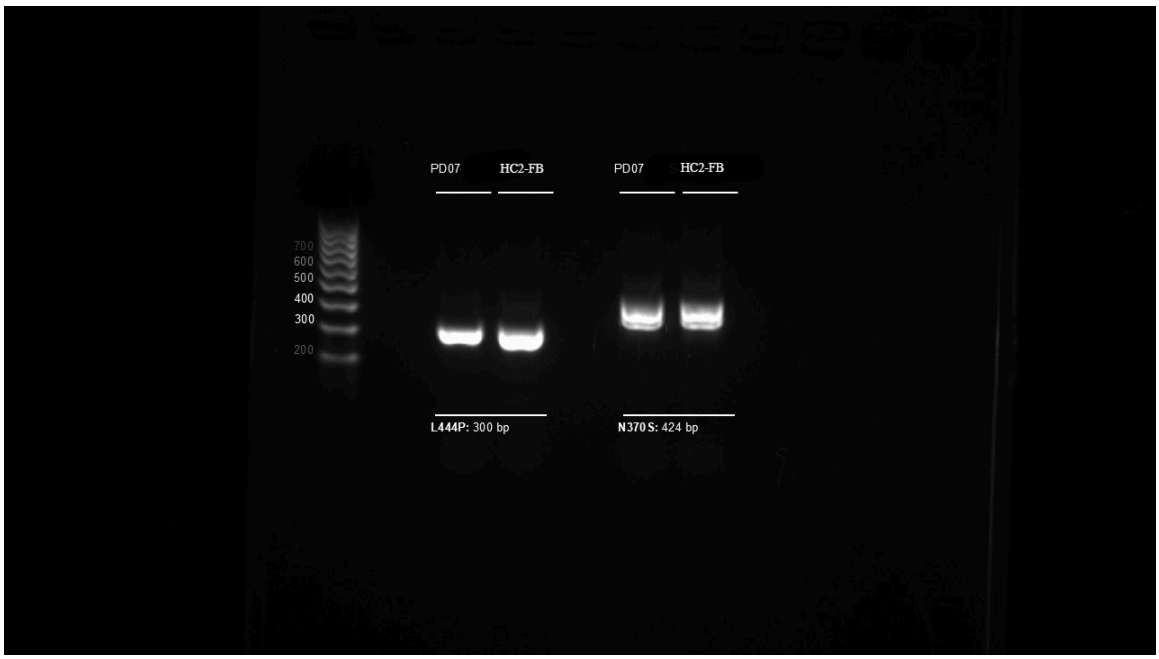
*Cell work:* Throughout the week, I fed and passaged the hiPSCs (PD05-eB01R11, eB01R42, mR01, and mR03) and fibroblasts (PD07) alongside Bin and Tatyana. Unfortunately, two of the lines did not survive the passaging Bin and I conducted on Wednesday: PD05-eB01R42 and PD05-mR01. On discussion, Bin, Tatyana, and I decided to test for mycoplasma and bacterial contamination. The BacT is negative as of Saturday (14Mar26). On Friday, I harvested PD05-eB01R11 for karyotyping, which is stored in Bin's box in the −80°C (label: `PD05-eB01R11 13Mar26`); I harvested mR03 (two vials: `PD05-mR03 P10 hiPSC 14Mar26`) on Saturday and passaged the PD07-FB at various seeding densities to begin reprogramming on Monday (16Mar26). The resulting cryovials were kept in the Mr. Frosty overnight and will be moved to the liquid nitrogen once we believe it is sufficiently safe for our existing stocks. These are detailed in the worksheets on my LabArchives.

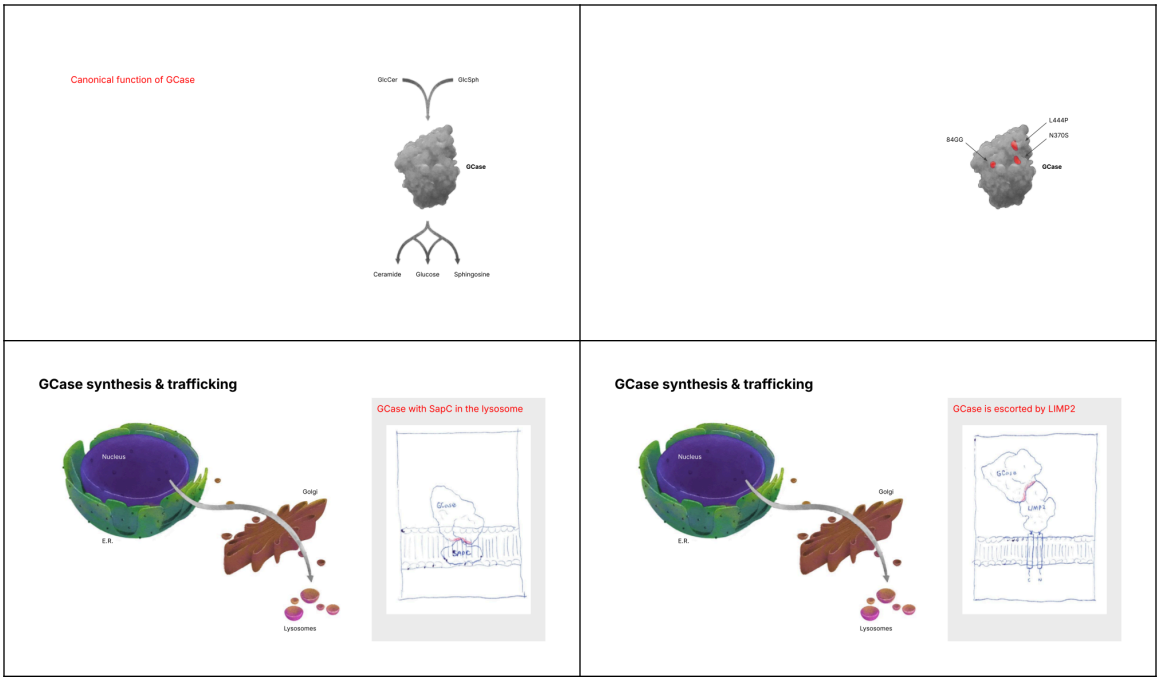
*Other TC preparation:* I prepared 12 40-mL aliquots of fresh NutriStem (Lot: 253102012, expiry 21Jan27) and stored them in the TC freezer. I created an updated inventory spreadsheet with a responsive calculator to determine working dilutions for TC reagents (`TC work > TC reagents and supplies > "Cell culture reagents for research.xlsx"` on Teams); and I created a (hopefully) more accurate template for printing CryoTag labels. I made up fresh 70% IPA and learned how to care for the TC equipment from Tatyana and Bin. I reviewed iPSC reprogramming protocols and began drafting an updated worksheet; I also copied and revised several existing worksheets, available on Teams with `"- HH"` appended to the file names.

### 3. *GBA1* presentation

Bin also asked me to prepare a ~30-minute presentation on *GBA1* and the planned battery of tests (see last update) for an upcoming lab meeting. I began preparing the slides, and also generated several figures (on my own time, after work) to assist in explaining *GBA1* biology. These are attached and are available to the lab for any needed use.

| Primer | Concentration (µM) | Stock conc. (µM) | Vol. primer (µL) | Vol. water (µL) |
|---|---|---|---|---|
| GBA1-L444P-forward | 249 | 10 | 20.08 | 479.92 |
| GBA1-L444P-reverse | 265 | 10 | 18.87 | 481.13 |
| GBA1-N370S-forward | 269 | 10 | 18.59 | 481.41 |
| GBA1-N370S-reverse | 260 | 10 | 19.23 | 480.77 |






*As you can see, these are still quite rough, and in some cases there are only my sketches in place. The first set demonstrates the function of GCase and the sites of our mutations; the second demonstrates its synthesis and trafficking (assisted by LIMP2) to the lysosome (where SAPC acts as a co-factor).*